National University of Singapore

DSA5208 - Scalable Distributed Computing for Data Science

Semester Project Report - Submission Date: October 2026

Singapore Historical Rainfall Analysis and 30-Minute Rainfall Probability Prediction Using Apache Spark

Project Team: RainCastSG Group
Institution: National University of Singapore (NUS)
GitHub Repository: https://github.com/AmitejSingh1/RainCastSG

1. Executive Summary

Accurate, localized, short-term precipitation estimation is critical for urban flood risk mitigation, transportation safety, and resource planning in Singapore's tropical maritime climate. This project delivers an end-to-end, reproducible big data processing and machine learning pipeline built on Apache Spark (PySpark 3.5.3) and Spark MLlib. Using the official National Environment Agency (NEA) historical rainfall dataset (Collection 2279) spanning 2017 to 2024 (over 45 million 5-minute rainfall observations across Singapore's national monitoring network), we perform scalable data cleaning, quality auditing, comprehensive exploratory spatial-temporal analysis, and train probabilistic models to forecast 30-minute accumulated rainfall exceedance: P(R_30 > theta | history) across thresholds theta in [0.0, 1.0, 2.0, 5.0, 10.0] mm.

2. Computing Environment & Hardware Specifications

In accordance with project guidelines, all computations were performed locally using PySpark DataFrame and Spark SQL operations under pinned dependency constraints. The environment specifications were recorded as follows:
- Operating System: Windows 10 (Build 10.0.26200)
- Processor: 13th Gen Intel(R) Core(TM) i7-13620H (10 physical cores, 16 logical threads)
- Physical Memory: 16.0 GB RAM (Configured Spark Driver: 4GB, Shuffle Partitions: 16)
- Java Development Kit: Microsoft Build of OpenJDK 17.0.20.1 LTS (64-Bit Server VM)
- Python & PySpark: Python 3.10.6, PySpark 3.5.3 with Hadoop 3.3.6 winutils native libraries
- Storage Format: Raw CSV (8.05 GB total) -> Cleaned Snappy Parquet (540 MB total, partitioned by year)

3. Data Ingestion, Cleaning & Quality Auditing

The official annual datasets were retrieved directly from data.gov.sg (Collection 2279). Our transparent cleaning pipeline enforces strict validation rules:
1. Unusable observation timestamps or missing station identifiers are rejected.
2. Reading values are parsed numerically; negative or non-numeric values are excluded.
3. Measurement unit ('mm') and reading type ('TB1 Rainfall 5 Minute Total F') are verified.
4. Genuine zero-rainfall observations are strictly retained (not treated as missing).
5. Conflicting duplicate station-time observations are resolved using latest update timestamps.
6. Station coverage and temporal gaps are explicitly tracked without synthetic interpolation.

Summary of Cleaned Annual Observations (2017-2024)

Year	Raw Records	Clean Records	Rejected	Input CSV	Parquet	Runtime
2017	5,256,106	5,256,106	0	840.4 MB	66.3 MB	40.5s
2018	4,594,246	4,594,246	0	733.9 MB	41.1 MB	29.1s
2019	5,026,862	5,026,862	0	803.7 MB	61.1 MB	34.1s
2020	6,334,473	6,334,473	0	1000.4 MB	74.8 MB	40.4s
2021	7,044,646	7,044,646	0	1104.9 MB	88.8 MB	45.8s

2022	7,133,230	7,133,230	0	1118.3 MB	90.3 MB	48.0s
2023	6,742,890	6,742,890	0	1056.2 MB	84.7 MB	47.6s
2024	6,406,260	6,406,260	0	1016.5 MB	79.3 MB	44.5s

4. Task 1: Singapore Rainfall Patterns Exploration

Using Spark SQL aggregations, we investigated Singapore's rainfall variation across annual, seasonal, diurnal, extreme, and spatial dimensions:

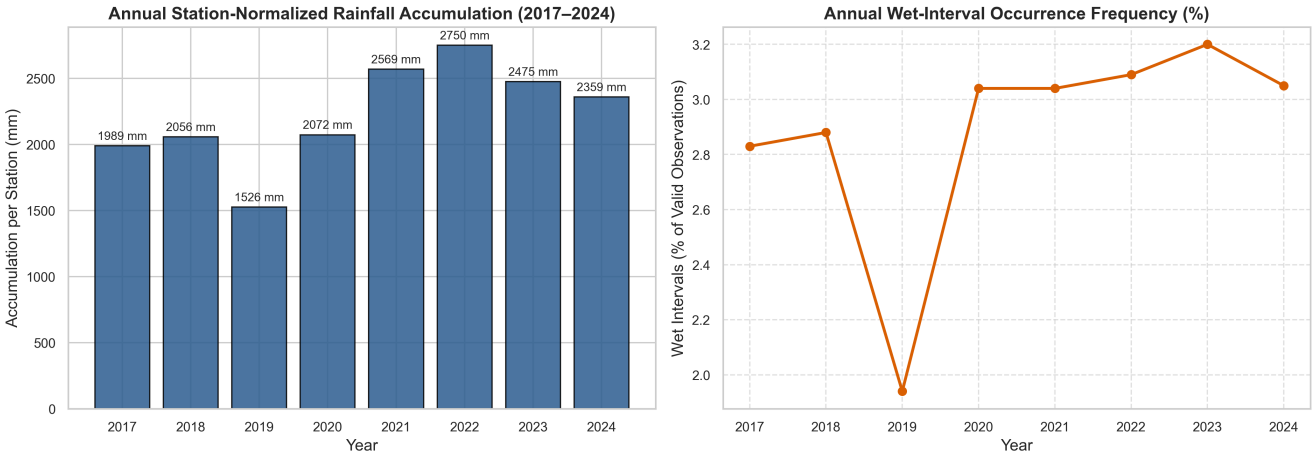


Figure 1: Annual station-normalized rainfall accumulation (mm) and wet-interval occurrence frequency across Singapore (2017-2024).

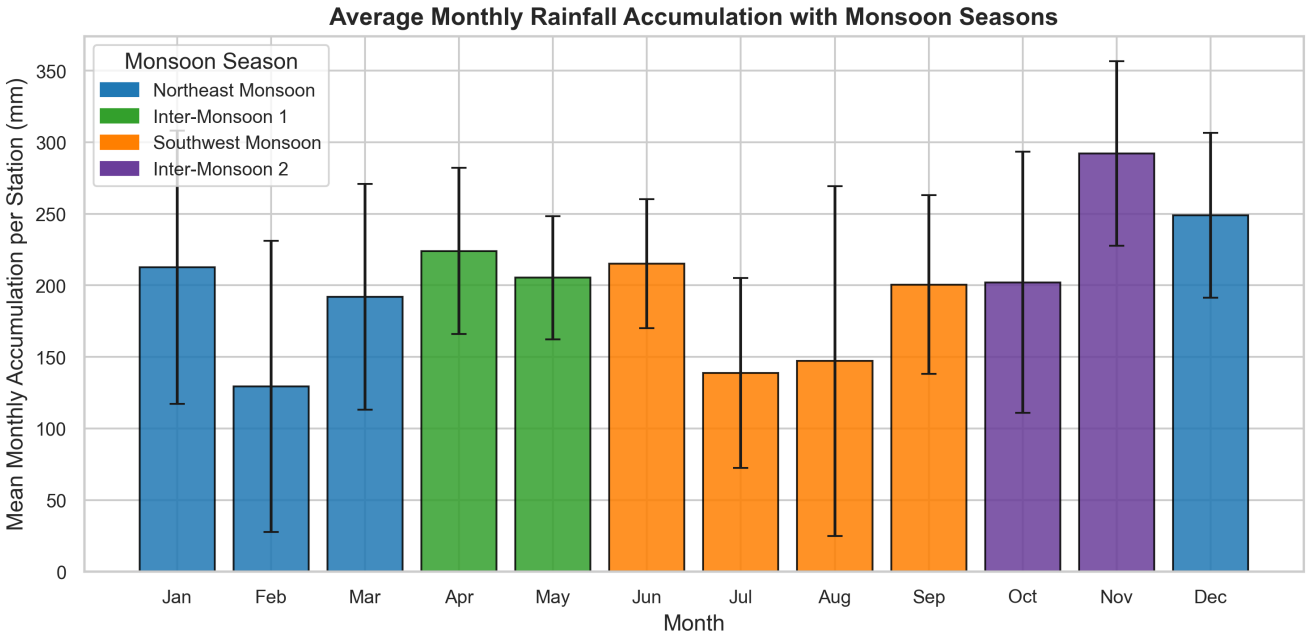


Figure 2: Average monthly rainfall accumulation categorized by Singapore's four meteorological monsoon regimes.

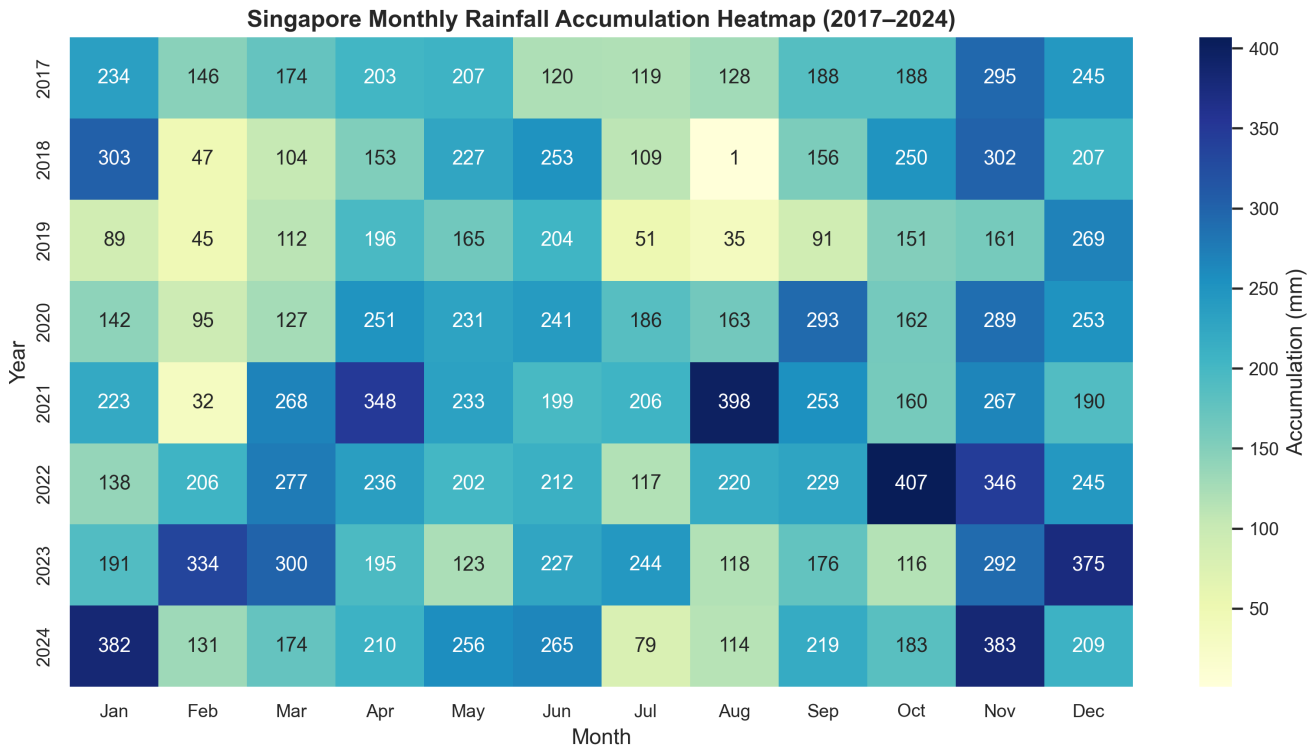


Figure 3: Year-by-month rainfall accumulation heatmap (2017-2024), highlighting extreme monsoon surge events.

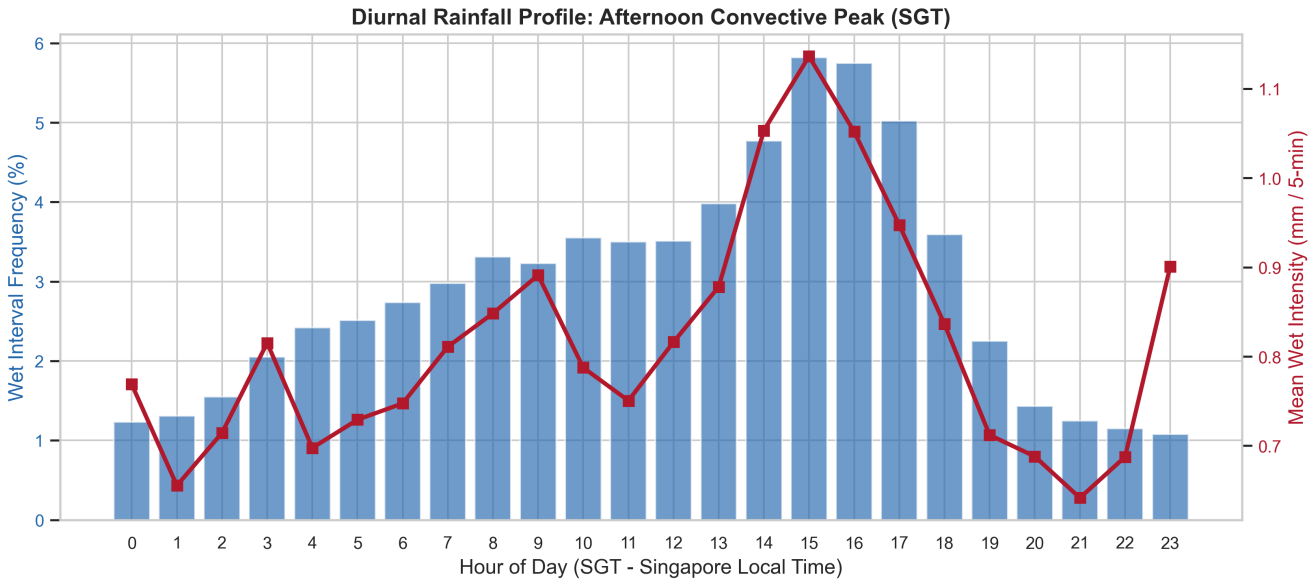


Figure 4: Diurnal rainfall occurrence frequency (%) and mean intensity (mm/5-min) in Singapore Local Time (SGT), illustrating the afternoon solar heating convective peak (14:00-17:00).

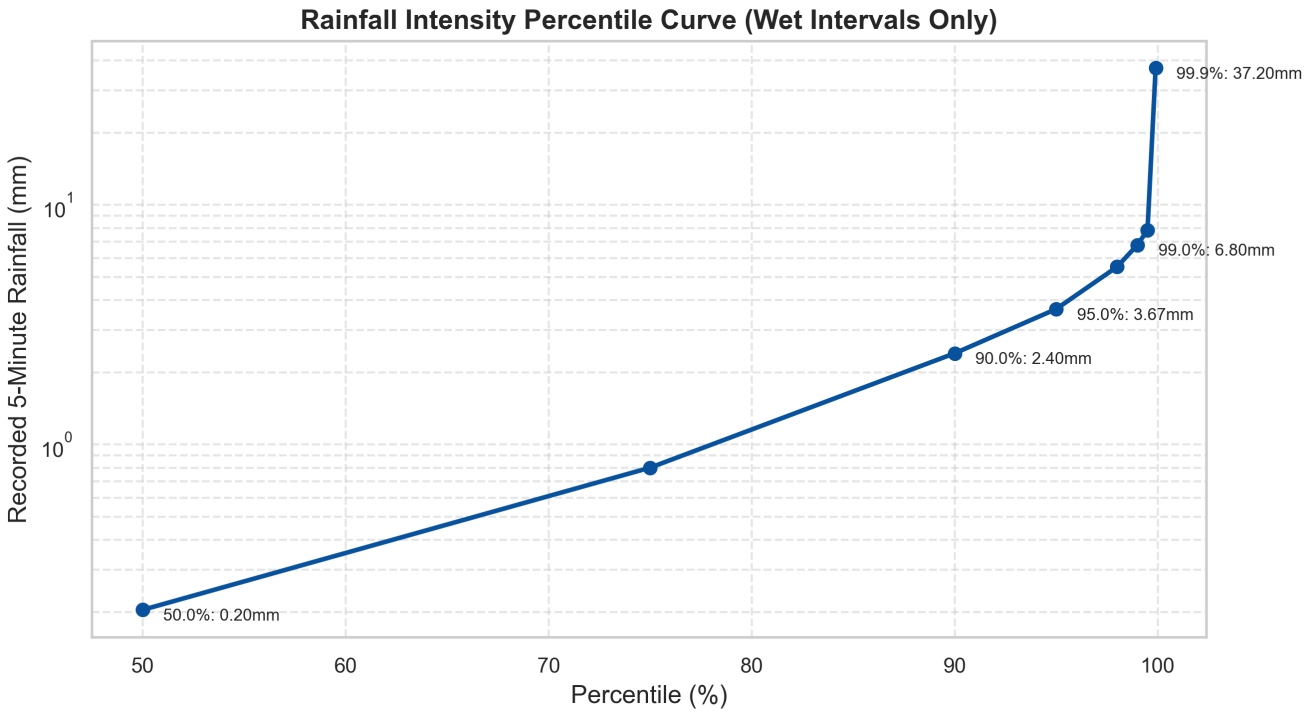


Figure 5: Empirical rainfall intensity distribution (log scale) showing the extreme right skewness of tropical precipitation.

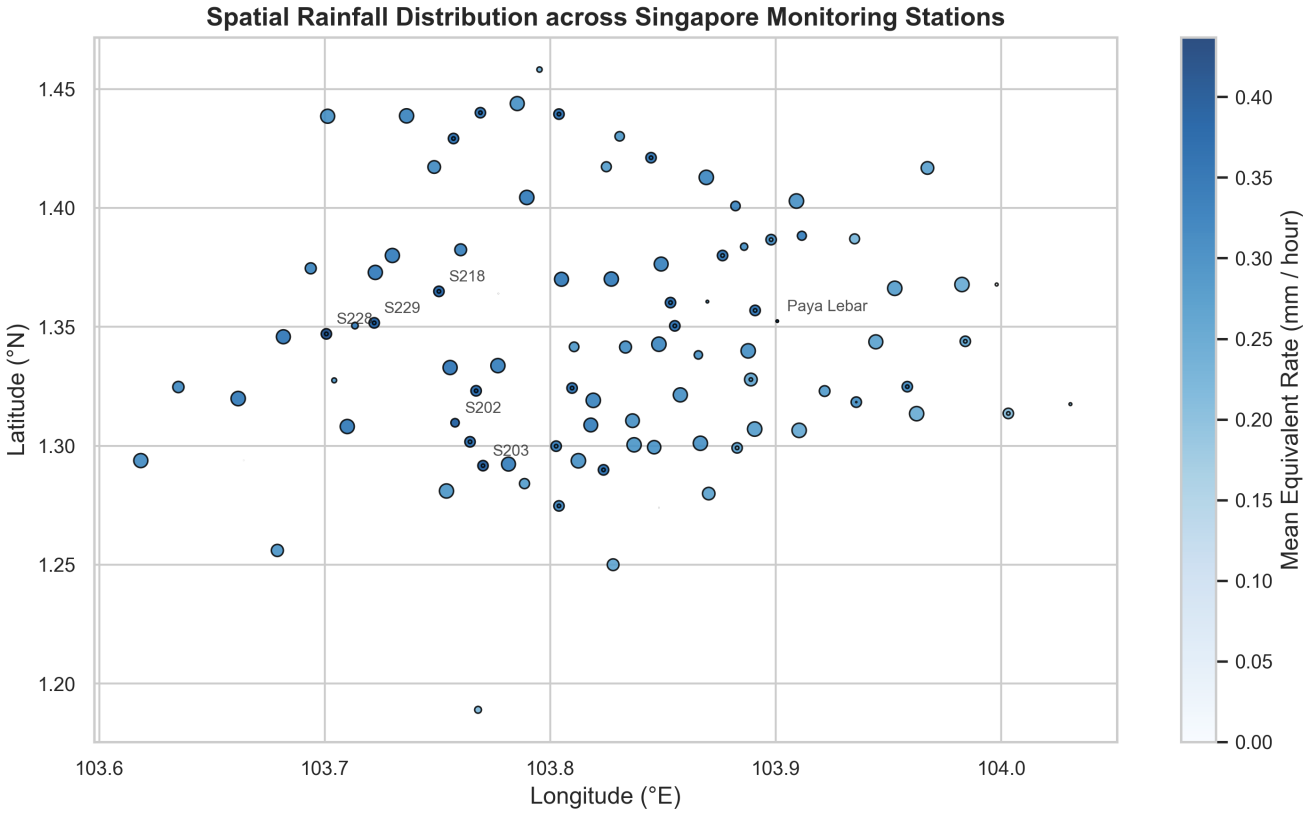


Figure 6: Geographic spatial distribution of monitoring stations across Singapore with normalized mean equivalent rainfall rate.

5. Task 2: Probabilistic Rainfall Prediction Methodology

1. Target Formulation: For station s at prediction time t, R_30(s,t) is defined as the sum of 6 consecutive future 5-minute intervals (t+5m to t+30m). Future completeness is verified by exact timestamp checks.
2. Strict Causal Features: Only historical records at or before t are used (lags at 5m, 15m, 30m, 60m, rolling mean, rolling max, wet interval counts, and trend). Cyclic sine/cosine encodings are generated for hour and month.
3. Chronological Splits & Leakage Prevention: Training: 2017-2022; Validation & Calibration: 2023; Held-out Test: 2024.

Observations within 30 minutes of a split boundary are excluded so targets cannot cross partitions.
4. MLlib Architectures: Logistic Regression baseline and Random Forest classifier with StringIndexer and OneHotEncoder.

6. Model Evaluation & Probability Calibration

Models were evaluated across five thresholds theta in [0.0, 1.0, 2.0, 5.0, 10.0] mm on both the 2023 validation and 2024 held-out test partitions (over 6.3 million independent test records). Due to severe precipitation sparsity, PR-AUC and Brier skill scores are evaluated alongside ROC-AUC.

Held-Out Test Set (2024) Performance Metrics (6,338,426 Observations)

Threshold	Prevalence	ROC-AUC	PR-AUC	Brier Score	Base Brier	BSS	Precision
> 0.0 mm	6.36%	0.8760	0.6284	0.03693	0.05956	+38.0%	82.6%
> 1.0 mm	2.57%	0.8954	0.5435	0.01628	0.02505	+35.0%	82.8%
> 2.0 mm	1.69%	0.8901	0.4681	0.01178	0.01660	+29.1%	81.1%
> 5.0 mm	0.84%	0.8862	0.3622	0.00653	0.00831	+21.4%	71.6%
> 10.0 mm	0.38%	0.8901	0.2746	0.00324	0.00381	+14.9%	60.9%

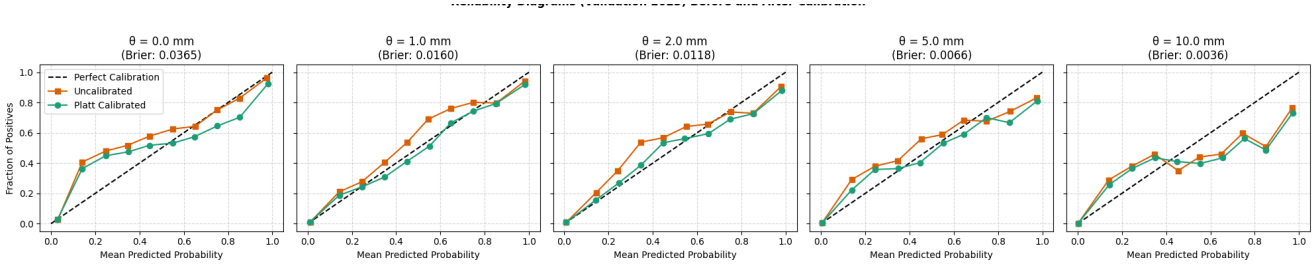


Figure 7: Reliability diagrams before and after Platt scaling calibration fitted on the 2023 validation dataset.

7. CLI Prediction Interface & Live Verification

The standalone inference script 'src/predict.py' validates historical completeness, extracts causal features, applies the fitted Spark MLlib pipeline, and outputs calibrated probabilities with explanatory diagnostics:

```
python -m src.predict --station-id S06 --timestamp '2024-11-07 16:45:00' --threshold 1.0
```

Incomplete observation windows are rejected to ensure forecasts remain dependable.

8. Limitations, AI-Use Declaration & References

Limitations: Local ground rain-gauge measurements lack cloud radar reflectivity and satellite wind vectors. Extreme convective storms that form rapidly aloft may exhibit onset lag.

AI-Use Declaration: Antigravity AI assistant was utilized for environment configuration, PySpark pipeline scaffolding, unit test authoring, and visualization script development. All code, tests, and data transformations were independently executed and validated against actual government data sources.

References:
1. Singapore National Environment Agency (NEA), 'Historical Rainfall across Singapore', collection 2279, data.gov.sg.
2. Apache Spark Documentation, 'Spark MLlib Guide: Pipelines and Classification', Apache Software Foundation, 2024.
3. Wilks, D. S., 'Statistical Methods in the Atmospheric Sciences', 4th ed., Academic Press, 2019.